

Academic Qualifications

Year	Degree/Certificate	Institute
2023 – 2026	BCA (AI & ML)	Galgotias University, Greater Noida
2023	CBSE (XII)	St. Gabriels School, Jabalpur
2020	CBSE (X)	Delhi Public School, Jabalpur

Technical Skills

- **Programming Languages:** Python
- **AI & ML:** LangChain, LangGraph, RAG, AI Agents, LLM Integration, Context Engineering, Traditional ML, NLP
- **Backend Development:** FastAPI, Flask, REST APIs, JWT Authentication
- **Databases ORM:** PostgreSQL, MySQL, Prisma ORM
- **Cloud DevOps:** Docker, GitHub, Google Cloud Platform (GCP), AWS

Work Experience

Backend Engineer – AI Infrastructure — MyPetAlly

PostgreSQL, Prisma, OpenRouter, Google Cloud Vision

- Designed & deployed a scalable AI backend **integrating OCR, LLMs & structured context** for pet healthcare insights.
- Built a **Multi-layer conversational pipeline** with memory management (5K tokens), streaming responses, and system guardrails.
- **Implemented** context compression **Reducing token usage by 42%** and optimizing LLM API **Cost Efficiency**.
- Developed **Emergency Detection** and validation logic with **Production-level reliability** and automated test coverage.

Key Projects

Multispectral Leaf Image analysis (Published in IJEDR, 2026)

 [Research Paper](#)  [GitHub](#) (Dec’25–Jul’26)

Objective	Developed a DL system for plant disease detection & severity estimation using RGB & multispectral Imaging .
Approach	• Trained Model-A (Multispectral CNN) on 5 spectral bands & vegetation indices for severity estimation. • Trained Model-B (RGB CNN) to classify whether a leaf is healthy or diseased from standard RGB images. • Implemented preprocessing pipeline including band stacking, normalization, resizing (128×128), and Performance evaluation using Accuracy, F1-score, ROC-AUC, & Confusion Matrix.
Results	• Demonstrated strong performance in disease detection and severity estimation, highlighting the effectiveness of combining RGB and multispectral deep learning models for precision agriculture.

Agentic AI Job Application Automation System (Independent Project)

 [GitHub](#) (Jan’26–Feb’26)

Objective	Developed an LLM-powered automation agent to streamline job applications by generating personalized emails, sending applications automatically, and tracking application status.
Approach	• Built a stateful LangGraph+LangChain workflow for application retrieval, email generation, delivery, & tracking. • Used OpenAI GPT models to generate personalized job application emails from job descriptions. • Connected with Google Sheets API & Gmail SMTP for automated email delivery with resume attachments. • Implemented automated scheduling using GitHub Actions, enabling continuous monitoring of pending applications and preventing duplicate submissions.
Results	• Automated the end-to-end job application workflow, reducing manual effort while maintaining personalized outreach and application tracking.

AI Data Analysis Agent (Independent Project)

 [GitHub](#) (Jan’25–Mar’25)

Objective	Developed an AI Agent capable of processing Multiple file formats & delivering interactive data analysis.
Approach	• Extracted data from diverse file types using pandas, docx, fitz, and PIL. • Achieved advanced language understanding & reasoning by implementing the LLaMA-4 API. • Generated insights & visual reports by Analyzing data, Answering queries & Handling follow-ups.
Results	• Delivered meaningful outputs by Extracting data, Analyzing content & Leveraging LLaMA-4 Maverick.

Extra-Curricular Activities

- Solved 150+ Leetcode Questions and participated in 7+ Hackathons.
- Selected as Interviewer for 2 internship candidates at MyPetAlly, assessing backend systems and AI fundamentals.
- Participated in Unifest’24, the North India’s Biggest Fest.